APPENDIX D — MATERIALS, SEALING, CLEANING, AND POWER/CHARGING (NON-LIMITING)

A. Overview

This appendix provides non-limiting examples of materials, sealing approaches, cleaning considerations, and power/charging configurations that may be used for any node described herein. Any material or subsystem described herein may be substituted with an equivalent while maintaining functional intent.

B. Materials and Biocompatible Interfaces (Non-Limiting)

In various embodiments, device housings and user-contact surfaces may be formed from one or more of: medical-grade silicone, silicone overmold on a rigid substrate, thermoplastic elastomers, polyurethane, polycarbonate, ABS, PEEK, nylon, or other suitable polymers. In certain embodiments, user-contact regions are configured as soft-touch and non-irritating surfaces. In some embodiments, user-contact regions are removable sleeves, removable liners, or replaceable consumables.

C. Sealing, Ingress Protection, and Fluid Resistance (Non-Limiting)

In various embodiments, nodes are configured as sealed assemblies to resist fluid ingress and enable cleaning. Sealing may include one or more of: overmolding, adhesive sealing, gasketed joints, ultrasonic welding, laser welding, potting of electronics, and sealed charge interfaces. In some embodiments, seams are minimized or eliminated in user-contact regions. In certain embodiments, the device is configured to meet an ingress protection target suitable for intended cleaning workflows.

D. Cleaning, Hygiene, and Storage (Non-Limiting)

In various embodiments, the device is configured for cleaning using one or more of: rinsing, wipe-down cleaning, mild soap cleaning, and/or approved disinfectant wipes, subject to material compatibility. In certain embodiments, the device includes a docking base, storage case, or drying cradle. In some embodiments, the device includes guidance elements indicating cleaning zones and non-cleaning zones. In some embodiments, the device includes removable accessories or sleeves to support hygiene and repeatable coupling.

E. Power Subsystems (Non-Limiting)

In various embodiments, nodes include a rechargeable battery, including lithium-polymer or other suitable chemistries. In certain embodiments, the device incorporates battery protection, thermal protection, and charge management. In some embodiments, the node is configured for power optimization using duty cycling, sensor burst modes, and event-triggered sampling.

F. Charging Interfaces and Docking (Non-Limiting)

In various embodiments, charging may be performed using one or more of: 1. inductive charging through a sealed housing, 2. magnetic pogo-pin docking with sealing features, 3. USB-based charging via sealed connectors, and/or 4. contactless docking with alignment magnets. In some embodiments, a docking base provides both charging and alignment for calibration or storage. In certain embodiments, the docking base includes indicators for charge state and/or pairing status.

G. Safety and Fault Handling (Non-Limiting)

In various embodiments, the system implements fault handling, including: low-battery behavior, overtemperature shutdown, charge fault detection, sensor fault detection, and memory fault handling. In certain embodiments, the system provides user-facing indicators for fault states, reduced-confidence states, and wear-detection states.

H. Variations

Any sealing approach, cleaning approach, material selection, and charging approach described herein may be combined with any other described approach. Certain embodiments emphasize fully sealed designs with inductive charging; other embodiments use docking with sealed contacts; other embodiments include replaceable sleeves or consumables.